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The role of livelihoods in agrifood sustainability transitions

Radhika Singh [a,*], Jampel Dell'Angelo [b], Nicholas Oguge [a], Collins Odote [a]

[a] Faculty of Law, University of Nairobi, Kenya
[b] Institute for Environmental Studies (IVM), Vrije Universiteit Amsterdam, the Netherlands

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ABSTRACT

Transitions to more sustainable agrifood systems are necessary to combat climate change and minimize environmental degradation. Such sustainability transitions have substantial implications for livelihoods, many of which will have to change to accommodate more sustainable practices. Livelihoods comprise the capabilities, assets, and activities individuals and households draw on for their means of living. However, transitions literature does not typically include analysis of how livelihoods are affected by, and affect, sustainability transition pathways. This limits understanding of factors that enable sustainability transitions in agrifood systems, particularly in the Global South. To address this gap, this article combines the Sustainable Livelihoods Framework (SLF) and the Multi-Level Perspective (MLP) into an integrated analytical framework, and applies it to the case study of rainwater harvesting for irrigation in Kenya. Results show that prevalent livelihood opportunities, assets, and strategies shape transition pathways by influencing levels of innovation adoption and scaling. First, agrifood sustainability transitions are more likely to take place if the innovation maximizes access to natural capital and creates new opportunities for households engaged in farm-based livelihoods. Second, such transitions can be accelerated when the adoption of a single innovation increases household access to multiple types of capital (social, physical, financial, human) at the regime-level. Third, the likelihood of such transitions taking place is affected by social and cultural institutions that influence household decision-making around the incorporation of innovations into livelihood strategies. These findings demonstrate that the inclusion of the sustainable livelihoods approach can improve understanding of the dynamics of agrifood sustainability transitions, and inform policies, programs or interventions aimed at guiding transition pathways.

1. Introduction

As demand for food is increasing, unsustainable methods of agricultural and livestock production are deepening environmental degradation and contributing to climate change (El Bilali et al., 2021). These agrifood systems, which organize activities and networks relating to the consumption and production of food, promulgate outcomes detrimental to ecosystem and human needs. Sustainability transitions to agrifood systems that produce better social and environmental outcomes are urgently needed (Matthews, 2004; Sietz and Van Dijk, 2015; Meynard et al., 2017). A "sustainability transition" is a type of sociotechnical transformation associated with specific sustainability targets and is a response to prevailing sustainability challenges (Feola, 2020). Sustainability transitions are often catalyzed by technological innovations that alter the dynamics of sociotechnical regimes, which are made up of dominant institutions

* Corresponding author.
E-mail address: rsingh10220@gmail.com (R. Singh).

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and technologies that direct developmental paths (Jenkins et al., 2018). While literature on sustainability transitions has helped identify and elaborate on processes involved in a transition, much of it was developed in the context of the Global North context and is euro-centric (e.g., Geels et al. 2002; Elzen et al. 2004; Geels and Raven 2006). Empirical cases are drawn from western democracies such as the Netherlands and Germany and grounded in their very specific types of economic, political and social systems. Findings from these cases may not always be applicable to sustainability transitions in other places around the world (Berkhout et al., 2015; Hansen et al., 2018; van Welie and Romijn, 2018; Feola, 2020).

The ultimate goal of systemic transformation can also be quite different between Global North and South countries (Feola, 2020). In the Global North, where high levels of production and consumption contribute disproportionately to climate change, sustainability goals are often aimed at achieving a low-carbon economy and focus on sectors such as energy or transport (Markard et al., 2015). In the Global South, on the other hand, sustainability goals are centered around how to achieve higher levels of well-being and address poverty challenges without over-exploiting natural resources (Scoones, 2015). Increasing emphasis on finding radical alternatives to capitalist-oriented development also means that institutions that are often taken for granted in transitions in Global North (e.g., the dominance of market mechanisms) may be questioned in other countries' sustainability transitions (Feola, 2020; Hebinck et al., 2021). However, generalizing differences between transition typologies in the Global North and in the Global South is undoubtedly an oversimplification. As it has been argued in the literature on the heterogeneity of capitalism(s), every state's role changes according to the specific historical and geographical context it is in (Fineschi, 2022; Hall and Soskice, 2001; Loewen, 2022). Since the state influences and shapes the institutional settings in which transitions take place, transitions are inherently political, and every country's transition dynamics, whether in the Global North or South, will have its own peculiarities.

Lessons in history, and indeed recent developments such as the Russia-Ukraine War, indicate that regimes in both the Global North and South can be rapidly destabilized and oscillate between stability and instability (Dell'Angelo et al., 2023). Nevertheless, it is possible to make a distinction between countries where regimes are more stable, less vulnerable to stresses, and can recover from disruptions more quickly, and countries where regime behavior is more uncertain, vulnerable to shocks, and recovery is not guaranteed. In the former, which includes many countries in the Global North, governments often have the ability to provide protection and support to niche innovations. While this makes it easier for innovations to scale, the overall stability of the regime means that systems are characterized by high levels of lock-in and path-dependence, and niche innovations may struggle to alter regime dynamics (Berkhout et al., 2015; Isgren and Ness, 2017; Vermunt et al., 2020). In the latter, which includes many countries from the Global South, the instability of dominant sociotechnical systems can open up more windows of opportunity to catalyze sustainability transitions. However, the unpredictability of the regime poses challenges to the scaling of niche innovations. In these contexts, actors in the regime may have to exhibit high levels of adaptive capacity to quickly respond to changes at the regime level and adopt relevant technological innovations (Edsand, 2019).

One of the most common ways individuals and households navigate the instability of dominant regimes, particularly if there is little support from social security programs that can cushion the effects of systemic disruptions, is to constantly adjust livelihood strategies (Scoones, 2015). Livelihoods comprise the capabilities, assets (both material and social resources) and activities for a means of living. In rural contexts in the Global South, livelihood strategies are often decided at the household level, and involve consideration of the effects of a change in one livelihood strategy has on the household's entire "basket" of livelihood strategies (Chambers and Conway, 1992; de Sherbinin et al., 2008; Djurfeldt, 2021). Livelihoods in the Global South tend to be part of the informal economy, pursued in combination with other livelihood strategies, provided little protection or support from the state, and are often temporary or short-term. The dynamism of these livelihood strategies and outcomes give rise to transition pathways that are often different from those in the Global North, where livelihoods are more stable and their influence on transition dynamics is more predictable. In such contexts, livelihoods can be treated as a fixed, long-term dependent variable when analyzing sustainability transitions.

For a transition to successfully promulgate more sustainable patterns of production and consumption, it needs to create reliable and low-risk livelihood opportunities that enhance households' socio-economic prospects while also circumscribing environmentally unsustainable outcomes. Agrifood systems transformation in particular has enormous implications for food security and income generation, as the majority of the population in the Global South is directly involved in agricultural production (Mwangi and Kariuki, 2015). In areas where the growing intensity and scale of agricultural production has contributed to soil degradation, water resource depletion, and ecosystem destruction, a transition to a more sustainable agrifood system depends on the creation of livelihoods that incorporate more sustainable agricultural practices and increase overall household assets. Sustainability transitions that do not prioritize support for these "sustainable livelihoods" can give rise to or exacerbate social, political, economic or environmental injustices and, in the long run, deteriorate support for sustainability transitions (Small, 2007; Scoones, 2015). Therefore, a stronger appreciation of livelihood dynamics, particularly in countries where households adopt a range of diverse, informal and short-term strategies in response to regime-level changes, can deepen understanding of how sustainability transitions are achieved.

Given the limited number of studies on sustainability transitions in the Global South, the role of livelihoods in reconfiguring relationships and shaping transition pathways is not very well understood. However, interest in the topic is rising in both the Global North and South, particularly in the context of enabling just transitions that can improve both livelihoods and environmental outcomes. Kaljonen et al. (2021), for example, emphasizes that food systems should deliver welfare and livelihoods while operating within sustainable environmental limits; Kuhmonen and Siltaoja (2022) assess how just sustainability transitions in Finland would have to create new livelihood opportunities for farmers; and Valverde et al. (2022) examine how gender gaps in Guatemalan family farming impact how just sustainability transitions are. In this paper, the role of livelihoods in a transition to a new agrifood regime in rural Kenya is investigated. This transition is aimed at making irrigation more environmentally sustainable and accessible to smallholder farmers through rainwater harvesting for irrigation (RWHI) systems, and involves the creation of more lucrative livelihood strategies.

To conduct this research, this paper combines two different but complementary bodies of literature: development studies with

particular attention to livelihoods, and transition studies that focus on long-term systemic change and improving sustainability outcomes. A novel framework that integrates the Sustainable Livelihoods Framework (SLF) with the Multi-Level Perspective (MLP) is developed to guide analysis. We apply this framework to examine how changing levels, combinations and sequencing of household assets give rise to new livelihood opportunities and catalyze niche formation around a particular technological innovation, and how innovation adoption reconfigures household assets to enable livelihood strategies that promulgate more environmentally sustainable and socially desirable outcomes. Our study provides a theoretical contribution by articulating how the livelihoods perspective can deepen understanding of transition dynamics and pathways, particularly in the Global South. The paper also proposes an original analytical approach that produces new empirical insights on agrifood sustainability transitions in Kenya. The following sections present a theoretical background that outlines the value of bringing the livelihoods concept into the literature of sustainability transitions, an analytical framework that combines the SLF and MLP and summarizes the study's methodology, findings from the empirical case of RWHI in Kenya, and a discussion of implications for sustainability transitions.

2. The need for a livelihoods perspective in the literature on sustainability transitions

Chambers and Conway (1992) characterize a sustainable (rural) livelihood at the household level as one that can cope with and recover from stresses and shocks, maintain or enhance its capabilities and assets, provide sustainable livelihood opportunities for the next generation, and contribute net benefits to other livelihoods at the local and global levels in the short and long term– all without undermining the natural resource base on which it depends. This definition emphasizes the role of the natural environment in shaping livelihoods and vice versa (Scoones, 2009, 2015; Vermunt et al., 2020). Incorporating the livelihoods perspective in analyses of sustainability transitions enhances understanding of conflicts between livelihood objectives that hinder sustainability efforts (for example, increased agricultural production for food security might deplete water resources that agriculture depends on). It helps determine how innovations fit into or are integrated into other livelihood strategies that a household engages in, and improves understanding how households or individuals perceive and act on tradeoffs of adopting an innovation. Integrating the livelihoods perspective can help assess effects of transitions on the overall sustainability of a household's "basket" of livelihood strategies, locate "lock-ins" in a sociotechnical system, and identify those livelihoods more resistant to change (Adato and Meinzen-Dick, 2002). For instance, the dependence of certain livelihoods on fossil fuel production can negate efforts to bring about a sustainability transition (Feola, 2020).

The livelihoods perspective is also relevant for assessing how sustainable a transition to a new agrifood regime actually is. Implied in the concept of sustainability transitions is the normative assumption that "sustainable" or "green" technological innovations will always yield positive social and environmental outcomes. The sustainability transitions literature is often more concerned with analyzing transition as a process rather than sustainability as an outcome. Yet the effects that technologies such as rainwater harvesting, solar panels or windmills can have on sustainability outcomes may differ significantly when they are translated into contexts outside of the Global North (Zhu et al., 2019; Feola, 2020). In many countries of the Global South, a rapidly changing society and environment has created uncertainties around how these innovations can contribute to the protection or maintenance of natural resources and what their wider effects on economic, political and social systems could be (Barrett et al., 2020). Using the lens of livelihoods can help assess answer questions such as– when is a transition sustainable, who is behind it, who benefits and how loses, and who is it sustainable for? (Köhler et al., 2019).

Furthermore, the literature on sustainability transitions has been criticized for not paying enough attention to the political economy of transitions (Avelino et al., 2016; WIlliams and Doyon, 2019; Geels, 2019; Hebinck et al., 2021). Diverse injustices, inequities, and dynamics of exclusion are part of any social system, but transition scholarship often assumes homogeneous experiences within communities and tends to use "civil society" to capture all actors not associated with either the state or the market (Avelino and Wittmayer, 2016). Without the explicit recognition that sustainability transitions have unintended and undesired consequences and create new vulnerabilities, the risk of either exacerbating or reproducing injustices that marginalize certain groups can increase (Hebinck et al., 2021). Incorporating the livelihoods perspective can help attune transition governance in the Global South to the particularities of interventions in informal, low-incomes settings that are characterized by high rates of poverty and inequality in order to ensure that transitions are as just as possible (van Welie and Romijn, 2018).

3. Analytical approach and methods

There are a number of analytical frameworks that have been applied in research on socio-environmental challenges and sustainable development pathways (Binder et al., 2013; Pulver et al., 2018). Previous contributions show how combining different analytical frameworks in novel elaborations can bridge separate but complementary bodies of literature and benefit from their different theoretical and methodological approaches (e.g., Clement 2010; McCord et al. 2017; Cole et al. 2019). In our opinion, SLF and MLP are among the frameworks that are most relevant, especially if combined, for understanding how livelihoods shape transition pathways in the Global South. Therefore, this paper proposes and employs an original integration of the SLF and MLP frameworks. The resulting framework provides a novel analytical approach for studying rural sustainability transitions in the Global South. The sections below elaborate on this integrated framework.

3.1. Integrating the SLF and MLP in a novel framework

The Sustainable Livelihoods Framework (SLF), developed by Scoones (1998), draws on the sustainable livelihoods concept and

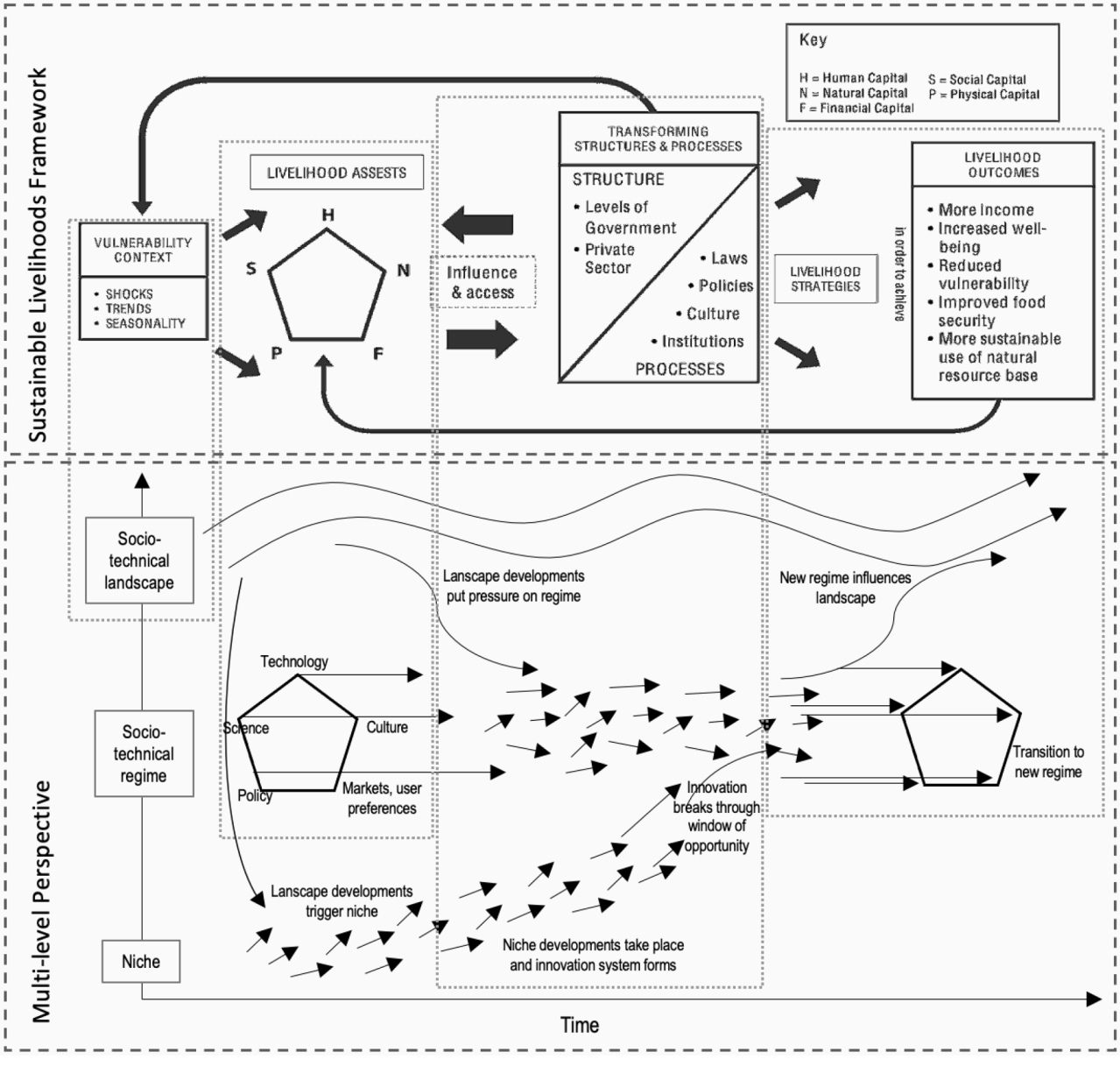

Fig. 1. Complementarities between the SLF and MLP (authors' elaboration based on Geels, 2006; Small, 2007; and El Bilali et al., 2017).

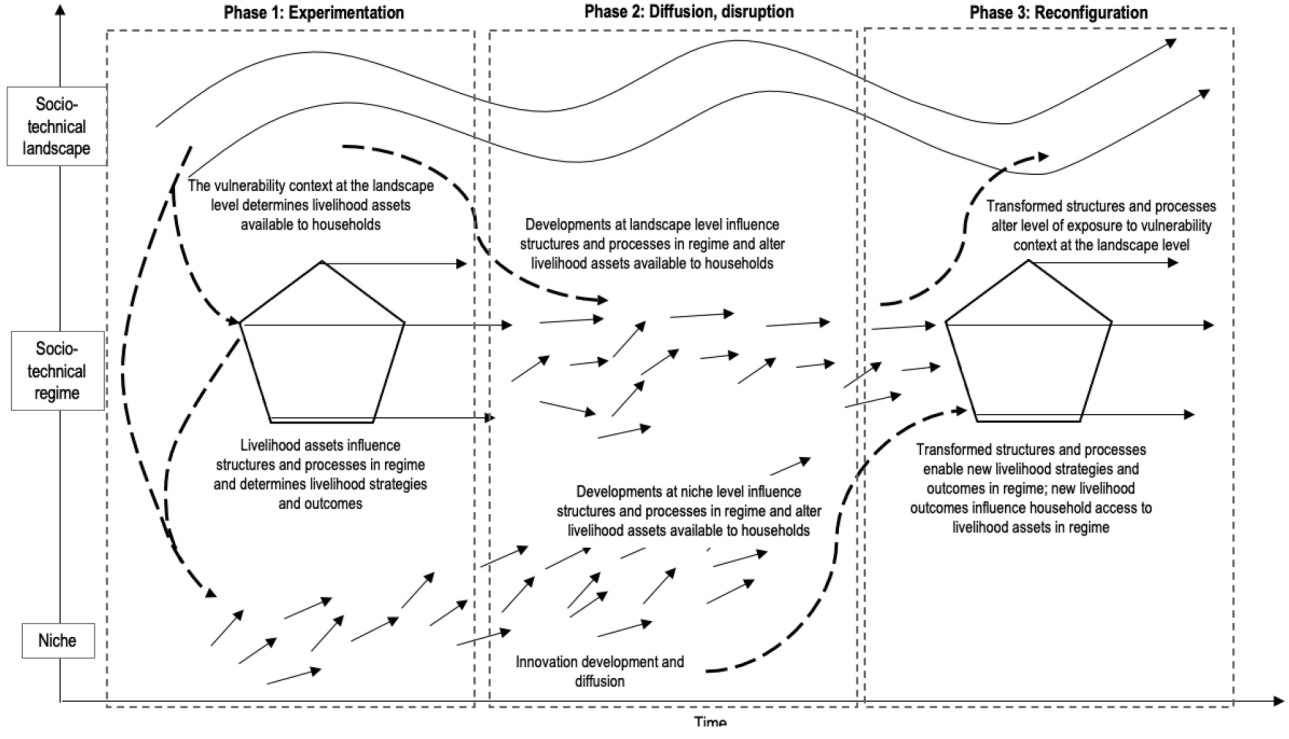

Fig. 2. Integrating the SLF into the MLP (authors' original elaboration).

enables development practitioners to think more holistically about levels, sequences and combinations of assets available to households and their effect on livelihoods (see Fig. 1). It is based on the idea that people require a range of assets to achieve positive livelihood outcomes, and that people's access, use, and development of these assets shape the sustainability of their livelihood strategies. The SLF, based on development concepts and values, is not formally linked to any one theory of change and is meant to be used in conjunction with other theoretical frameworks (Small, 2007; Scoones, 2009; Levine, 2014; Biggs et al., 2015; Scoones, 2015; El Bilali et al., 2017; Natarajan et al., 2022).

The SLF enables analysis of the links between the vulnerability context, transforming processes and structures, and livelihood assets, strategies and outcomes. The vulnerability context consists of exogenous elements (e.g., macroeconomic factors, international economic disruptions, climate-related shocks) that impact assets available to households. Transforming processes and structures consist of changes in dominant systems' formal or informal institutions that also alter the assets that households have access to. These livelihood assets include a. human capital, which represents skills, knowledge and good health; b. social capital, which includes belonging to networks and groups, and maintaining other relationships of trust, reciprocity and exchanges; c. natural capital, which represents natural resource stocks such as land, ecosystems, water bodies, etc.; d. physical capital, which comprises of basic infrastructure needed to support livelihoods; and e. financial capital, which represents cash or any equivalent gained through savings, investments or income. Livelihood strategies, which can be understood as combinations of activities that people undertake to achieve certain livelihood outcomes, arise from household's access to certain levels, combinations and sequences of these assets. Livelihood outcomes are the ultimate reasons for why people engage in livelihoods, although the relative importance of certain outcomes may change depending on the context (Alinovi et al., 2010; Reed et al., 2013; Nyamwena-Mukonza, 2013; Scoones, 2015; Serrat, 2017).

Deriving from the literature on sustainability transitions, the MLP captures the long-term, multi-scalar transformation processes embedded in sustainability transitions and provides a framework for analyzing interactions between the macro-level landscape, meso-level regime, micro-level niche (see Fig. 1). It is used to understand how innovations break out of protected niche spaces to catalyze system transformation. In the MLP, the "landscape" consists of economic, political, social, cultural, and environmental factors that shape the development of an innovation system but that are not themselves directly affected by it. This landscape influences both the "niche", which consists of incubation spaces for specific technologies that are created by actors and supported by specific institutions, and "sociotechnical regimes", which are made up of coherent, interrelated and stable structures of established actors, institutions and networks around a technology. Regime changes can be triggered by landscape pressures, such as pandemics, and/or developments at the niche-level, such as the development of new technologies. When innovations are adopted and scaled, moving from niche to regime, they can alter regime dynamics and engender more sustainable systemic outcomes. By making explicit the interplay between stabilizing regime mechanisms and destabilizing landscape and niche pressures, the MLP helps explain innovation and scaling processes involved in sustainability transitions (Geels and Schot, 2007; Markard and Truffer, 2008; Alinovi et al., 2010).

The SLF and MLP were developed for different reasons and are usually applied in different contexts. They have not been applied in an integrated way and the implications of their integration have not been fully explored. As far as these authors know, El Bilali et al.

Table 1
Complementarities between SLF and MLP in different phases of a sustainability transition.

Phases	Phase 1: Experimentation	Phase 2: Diffusion, disruption	Phase 3: Reconfiguration
Multi-level Perspective	Landscape developments put pressure on regime, leading to the creation of niche spaces for innovation development	Landscape developments destabilize regime and create "window of opportunity" for radical innovation to break through	New infrastructure, political institutions, and actor-networks reconfigure regime, affecting landscape-level dynamics
	Actor-networks pioneer radical innovations in niche; learning processes take place amidst high degree of uncertainty around innovation potential	Radical innovation breaks out of protected space in niche and establishes a foothold in market niches in regime and begins to move in mainstream markets	New technologies replace existing ones in regime and promulgate new socio-technical and environmental outcomes
	Regime for markets/user preferences, culture, technology, science, and policy is locked in; system elements change incrementally making it hard for radical innovations to break through	As innovation's trajectory begins to stabilize and form its own trajectory, it competes with institutions in existing regime. Struggles take place between actor-networks in incumbent regime and those supporting niche innovation	Altered regime develops own patterns of path-dependence and lock-in, culminating in a sustainability transition. Regime is one again more inclined to incremental rather than radical change
Sustainable Livelihoods Framework	The vulnerability context determines levels, combinations, and sequences of livelihood assets available to households	Changes in transforming structures and processes influence levels, combinations, and sequences of livelihood assets available to households	Transformed structures and processes enable new livelihood strategies and outcomes in regime; new livelihood outcomes influence household access to livelihood assets in regime
	Livelihood assets that result from the vulnerability context influences prevalent structures and processes in society		Transformed structures and processes and altered level of exposure to vulnerability context
	Structures and processes determine potential livelihood strategies and outcomes		

(2017) is the only scholarship that elaborates on synergies between the SLF and MLP, highlighting complementary concepts between them and developing a framework aimed at examining rural livelihood diversification endeavors. In Fig. 1, which draws from Geels (2006) for the MLP; Small (2007) for the SLF, and El Bilali et al. (2017) for the synergies between the two frameworks, complementaries between the SLP and MLP are illustrated through boxes grouping together concepts from each framework. While El Bilali et al. (2017) present the broad strokes for the complementaries and potential of integration between SLF and MLP, their conference paper does not investigate the implications of this integration on assessments of sustainability transition pathways. As a novel contribution, this article integrates the two frameworks to deepen understanding of the role of livelihoods in enabling or constraining sustainability transitions and outcomes.

Complementing the MLP with the SLF allows one to "zoom in" on elements most relevant to livelihood dynamics at the niche, regime or landscape levels. While MLP is dynamic and structure-oriented at a regional level, the SLF is relatively static and agency-oriented at the household level. Their integration enables assessment of how livelihoods in dominant socio-technical systems shape transition pathways by driving the creation of niche spaces, giving rise to "windows of opportunity" for innovations to scale, and contributing to new social and environmental outcomes. To make the complementarities between the MLP and SLF clearer, we have elaborated on conceptual synergies between the three phases of a sustainability transition: experimentation, diffusion/disruption and reconfiguration (adapted from Geels and Turnheim 2022) in Fig. 2 and Table 1. The landscape in MLP includes the vulnerability context concept in SLF. Landscape- and niche-level developments affect the sociotechnical regime, which is made up of households leveraging available assets for their livelihood strategies. These developments, which open up windows of opportunity for innovations in the MLP, are represented by the SLF's transforming structures and processes concept. When innovations begin to alter levels, combinations and sequences of household assets at the regime-level, new livelihood opportunities are created, triggering new livelihood strategies. This can ultimately give rise to novel livelihood outcomes that catalyze shifts towards more sustainable regimes.

As Table 1 indicates, the spatial and temporal boundaries of the two frameworks are different from one another. The MLP seeks to conceptualize the long-term system dynamics involved bringing about more sustainable socio-technical regimes. Often, this process can take decades, making the MLP an outcome-oriented framework that can aid in development of multi-year transition pathways or in the retrospective assessment of a particular transition. System boundaries are drawn by experts in a top-down manner based on what the goal of their analysis is, and tend to use indicators widely accepted by the scientific community. The functioning of institutions is analyzed at the regional level, generally focusing on more formal policies, practices, and regulations. However, due to the level of reductionism necessary to capture such long-term processes and present generalizable implications, the MLP does not emphasize the complex range of innovation user perspectives and behaviors. The SLF, on the other hand, presents as a "snapshot" of a household's or community's livelihood opportunities, strategies and outcomes in a specific context and temporal period. It offers a contextualized understanding of local factors affecting household- or community-level dynamics, and focuses on household or community level institutions that encompass informal socio-cultural tendencies or preferences. The more bottom-up perspective of the SLF is often based on indicators that emerge through community engagement, and findings stemming from applying SLF are often less generalizable outside the particular context (Reed, 2006; Saxena et al., 2016).

Combining the MLP with the SLF can provide an integrative method to bring together multiscalar, longitudinal, and generalizable insights on sustainability transitions, and specific, time-bound insights on local livelihood dynamics in chosen context. Apart from the MLP, the SLF can also be applied to frameworks from the literature on the diffusion of technological innovations, which looks more narrowly at how an innovation can break out of a niche and enter a regime. However, the MLP was chosen because it can aid analysis of how innovations can alter livelihood dynamics and reconfigure systemic behaviors at the regime level. Moreover, it has broad acceptance in the sustainability transitions research as is flexible enough to be adapted to various contexts, including those of the Global South. Indeed, the body of literature that applies the MLP to sustainability transitions in the Global South is only growing. However, it is important to recognize limitations to the application of MLP to the Global South context. Van Welie et al. (2018), for example, argue that the MLP does not take the heterogeneity and unevenness characteristic of socio-technical systems in developing cities. Integrating the SLF into the MLP can help overcoming the limits of the MLP's applicability to the Global South by enhancing the analysis of complex and changeable multi-layered regime dynamics through the livelihoods perspective.

3.2. Using the framework to deepen insights of sustainability transitions

Combining the SLF with the MLP into an integrated framework enables assessment of how: a. landscape level developments determine the degree of household access to livelihood assets at the regime level; b. access and use of available livelihood assets by households at the regime level gives rise to problems or opportunities that create niche spaces for innovation development; c. the adoption of a niche innovation alters combinations, sequences and access to household livelihood assets at the regime level and creates new livelihood opportunities; and d. how households take advantage of new livelihood opportunities and engage in new livelihood strategies, leading to new systemic dynamics and sustainability outcomes.

These assessments can deepen insight into the types of transition pathways most conducive to achieving desired social and environmental outcomes. Although such outcomes are often assumed in analysis of sustainability transitions, undesired consequences are part of any intervention in a complex system, and transition pathways must be adjusted accordingly. However, it can be difficult to identify and mitigate these consequences without some way of qualifying the "sustainability" in sustainability transitions. With the explicit inclusion of "natural capital" and the implicit incorporation of environmental factors in the SLF's "vulnerability context" and "transformative processes and structures", as well as the emphasis on sustainable natural resource management in the definition of sustainable livelihoods, the analytical framework integrating the SLF into the MLP helps assess the effects of a transition on the natural environment (e.g., ecosystems, water resources, climate, etc.). By connecting causes (the introduction of an innovation in a niche) and

outcomes (sociotechnical change) through its focus on livelihoods, it also provides a lens to assess how far a transition actually goes towards meeting both social and environmental sustainable development goals in the long run.

Furthermore, the framework enables analysis of how households navigate cross-scale dynamics between coupled social, technical, and ecological systems in different transition pathways (Hebinck et al., 2021). In sustainability transitions, innovations can dramatically change the distribution of and access to capital assets, potentially destroying some livelihoods and creating new livelihood opportunities (Adato and Meinzen-Dick, 2002). Integrating the SLF into the MLP directs attention to how an innovation alters the way household use and sequence assets available to them, how these assets interact with each other, what effect these interactions have on the speed or nature of transition pathways. For example, social capital could become a way of overcoming constraints in natural capital (collective water management); limitations in human capital could decrease financial capital (limited labor availability discourages planting of more labor-intensive maize varieties); or physical capital could enhance potential of natural capital (agricultural technologies maximize land potential). The framework can shed light on the tradeoffs households make when adopting new technologies and help identify assemblages of livelihoods, innovations and politics that may generate more sustainable futures (Scoones, 2015).

The framework could also enhance analysis of the political economy of sustainability transitions. Although the MLP has been used to examine power dynamics, studies have tended to focus on dynamics between different scales (niche, regime, and landscape) or formal actors (firms and governments). However, transitions are also shaped by intra-household or intra-community dynamics and directed by powerful, informal actors (Avelino, 2017; Köhler et al., 2019) whose actions can deepen pre-existing injustices and inequalities that worsen patterns of marginalization and inequity (Isgren and Ness, 2017; Wang and Lo, 2021). Incorporating the SLF into the MLP can therefore help answer questions such as: How are abundancies or scarcities created during a transition, by whom, and to what ends? Who owns what (access to assets), who does what (social divisions and roles), who gets what (patterns of accumulation), and what do they do with it (livelihood strategies, patterns of consumption, social reproduction)? (Bernstein, 2010; Scoones, 2015; Köhler et al., 2019; Sovacool et al., 2021; Wang and Lo, 2021). The SLF highlights that in a just transition, stresses and shocks must be recovered from, assets must be maintained and enhanced, and the natural resource base should not be undermined (Scoones, 2015). This political economy perspective emphasizes that transition pathways, particularly for agrifood regimes, are highly context-specific and place-based (Vermunt et al., 2020).

3.3. Case study and study sites

To deepen understanding of the role of livelihoods in sustainability transitions, this research analyzes the case of RWHI. RWHI is a technological innovation that has potential for improving sustainability outcomes in agrifood systems and improving the livelihoods of smallholder farmers in sub-Saharan Africa. It can enhance food and water security and enable income-generating activities without contributing to the over-abstraction of surface water resources. By using harvested rainfall during drier months, households can grow enough food for subsistence or commercial purposes and store enough water for domestic and livestock needs (Enfors, 2013; Bunclark, 2018; Piemontese et al., 2020). RWHI is therefore a relevant case to examine how innovations catalyze sustainability transitions through their effect on livelihoods. There are many types of RWHI, but our study focuses on *ex-situ*, meso-catchment RWHI in the form of household-level farm ponds that collect run-off from nearby roads or paths after rainfall, as this type of system has been shown to have the most potential to provide enough water for irrigation compared to other types of RWHI (Gomez et al., 2018). For the rest of this article, the term RWHI will refer only to farm ponds, which often involve the use pumps, drip lines or sprinklers to abstract and distribute water, and dam liners and shade nets to maximize water storage capacity.

The case of RWHI is examined in Kenya, a country that serves as a relevant context to ground analysis of the link between sustainable livelihoods and sustainability transitions. As a lower-middle income country in the Global South, Kenya is experiencing rapid agricultural development and known for being a center for agro-innovation development and diffusion, with major effects on livelihoods (Goldsmith, 2016). Moreover, Kenya has undergone an important water governance transition from a monocentric, top-down system to one exhibiting traits of polycentricity, which has implications for sustainability outcomes (Baldwin et al., 2016; Dell'Angelo et al., 2016; McCord et al., 2017; Baldwin et al., 2018). Initial interviews with experts in this field, site visits and non-participant observation were conducted to determine the most relevant sites for this research. Laikipia Plateau and Yatta Plateau were chosen, since a substantial amount of agriculture-related research has been carried out in these regions and they have a long history of well-documented RWHI interventions beginning in the 1980s.

Laikipia Plateau is located primarily in Laikipia country, while the Yatta Plateau is located in both Machakos and Kitui countries. Both plateaus have a semi-arid climate with frequent dry spells, although Laikipia benefits from more surface water resources due to its location on the leeward side of Mt. Kenya (Ngigi et al., 2008; Mburu et al., 2015). Both sites have undergone major social, political, economic and environmental transformations over the last sixty years, including land reforms and in-migration from the 1950s to 1980s, structural adjustment policies prioritizing export-led agricultural development in the 1980s and 1990s, and constitutional reforms resulting in the devolution of political power to newly formed counties in 2010 (Recha et al., 2018). The wealth of information available on the sites allowed for a deep understanding of how households' livelihood assets, opportunities, strategies and outcomes, as well as the broader vulnerability context and transforming processes and structures, have contributed to a sustainability transition (Ulrich et al., 2012; Mburu et al., 2015). However, ascertaining the exact extent of RWHI adoption in each of the two sites is challenging for a number of reasons. For one, adoption is often a dynamic process, with periods of increased engagement with the innovation and periods of withdrawal or disengagement (Bunclark, 2018). Assessment of the rate of adoption at a particular period of time cannot be conceived of a point on a linear scaling trajectory and therefore may not accurately indicate past or future rates of adoption (Wigboldus et al., 2016). Secondly, the largely informal nature of smallholder agriculture means that farmers adopting

irrigation systems, including RWHI, are not officially on record at the county or national level (Mati, 2023). Finally, while satellite data has been used to identify households with RWHI, seasonal changes in water levels and visual blockages such as trees and vegetation require extensive ground-truthing to verify areas of adoption (Malesu, 2007).

Given these limitations, as well as the researcher's own time and resource constraints, a general sense of the levels of adoption was first gained through detailed analysis of all relevant studies and reports on RWHI in the Laikipia and Yatta Plateaus. These studies demonstrated that RWHI technologies, including dam liners, drip lines or sprinklers, and diesel or solar pumps, were adopted at a much higher rate in Laikipia compared to Yatta. For example, Wachira (2013), in his study in Matanya, Laikipia, demonstrated that RWH was the most important source of irrigation for 45 % of households. Confirming this, Mekdaschi and Liniger (2013) found that the diversification of water sources for irrigation and increased adoption of RWHI has significantly decreased farmer dependence on river water, especially during the dry season. Lanari et al. (2018) further indicate that the use of river water for horticultural irrigation in Laikipia had decreased by 31–35 %, while the use of harvested water increased by 14–21 % between 2003 and 2013. In Yatta, on the other hand, studies have found that between 1.6 % (Ahmed et al., 2013) and 2 % (Mburu et al., 2015) of households have adopted RWHI. Water resources are scarce and water for domestic consumption and livestock is prioritized over other uses. Therefore, farming is largely rainfed and utilizes mostly traditional tools (Makau et al., 2014). These findings and their alignment with historical patterns of RWHI adoption were confirmed through interviews and focus group discussions with agricultural extension agents, agricultural input businesses, horticultural companies, and NGOs located in Laikipia and Yatta, many of which have been operating in the area for decades.

In this study of RWHI in Kenya, two agrifood regimes– one of the Laikipia Plateau, and the other of the Yatta Plateau– were used to frame this analysis. These agrifood regimes comprise of existing practices, technologies, and governance structures that govern how food is produced, processed, and distributed. Several aspects of each agrifood regime influence the potential for RWHI scaling. Governance mechanisms and dynamics at county levels, as well as national-level policy and legislation, can either enable or constrain the scaling of RWHI. Financial and economic structures within the agrifood regime determine access to credit and investment capital and households' ability to invest in innovative irrigation technologies. Moreover, the existing input and output market structures may not provide sufficient incentives for adopting RWHI, especially if the benefits of such systems are not immediately tangible or financially rewarding. Finally, traditional farming practices, social norms, and cultural beliefs can either facilitate or impede the adoption of RWHI. In both Laikipia and Yatta, RWHI can be understood as moving from a protected niche space to the dominant agrifood regime when externally funded projects are no longer the primary catalyst for adoption. Where RWHI systems are still being initiated, financed, and guided largely by government or NGO projects, then RWHI is still in its niche and adoption is rarely sustained when support is removed. RWHI has entered the regime when RWHI systems are established and sustained mainly through self-financing by households, a strong private sector providing a diverse range of RWHI technologies is present, and other socio-technical systems in the regime begin to rely on the presence of RWHI systems for their own functionality (see Table 2).

3.4. Data collection and analysis

Primary data was gathered in the form of unstructured and semi-structured interviews and focus group discussions with relevant stakeholders, as well as through information supplied in official policy documents. Secondary sources included both scientific articles and NGO project reports. In an iterative process, existing secondary data was collected during and between phases of primary data collection to establish the context of rural agricultural livelihoods and fill in information gaps. This process continued until saturation,

Table 2
Dimensions, actors, and processes at each MLP level from the SLF perspective.

	MLP: Levels and scales	SLF: Dimensions, actors, and processes	Application to case study of RWHI in Kenya
Landscape	The landscape level in MLP describes slow-changing states including demographics, social values, culture, macro-economic trends, and external shocks (pandemics, trade wars, geo-political developments, etc.)	The vulnerability context in the SLF homes in exposure to shocks, trends and seasonal changes in the landscape level that determine availability of and access to livelihood assets and have the potential to transform structures and processes at the regime level.	Vulnerability context at the landscape level includes history of marginalization under colonial rule; migration and land-use changes; water and land scarcity
Regime	The socio-technical regime in MLP is made up of existing socio-technical systems that are stabilized by lock-in mechanisms; technological, scientific, political, cultural and market institutions move along path-dependent trajectories.	The livelihood assets concept in SLF encompasses the types of human, social, natural, physical and financial capitals available to households in a specific socio-technical system at the regime level. These livelihood assets shape structures and processes that drive system trajectories and give rise to certain livelihood strategies and outcomes at the regime level.	Livelihood assets in socio-technical systems relevant to RWHI include knowledge of irrigation; social institutions to manage water use; enough rainfall to harvest water; infrastructure to access input and output markets; financial capital to make investments in RWHI technologies.
Niche	The niche in MLP is where radical innovations emerge in protected spaces, until they ultimately break through to the regime level.	Developments in the niche transform structures and processes at the regime level, altering livelihood assets available to households.	Transforming processes and structures include development of RWHI technologies in a niche space by NGOs for household water security and by the private sector for commercial farms.

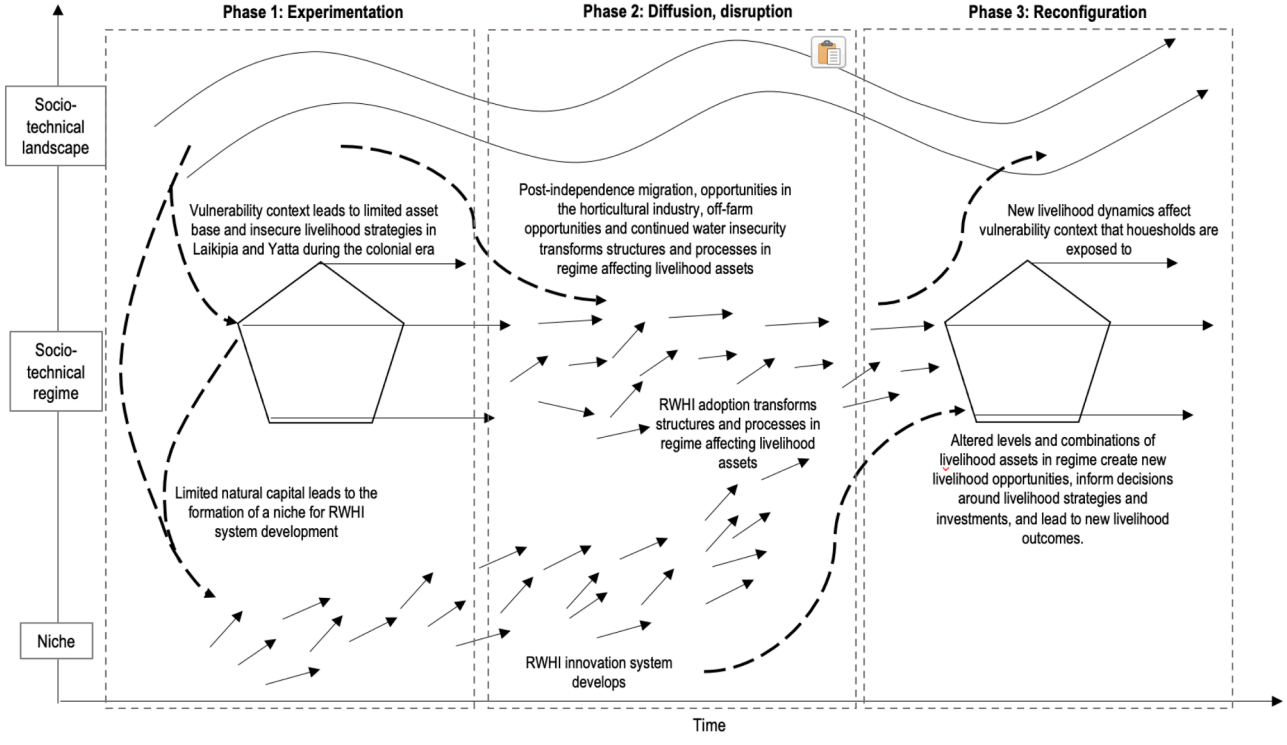

Fig. 3. Integrating the SLF into the MLP for the case of RWHI in Kenya (author's elaboration).

or the point at which iteration between theory and data became minimal, was achieved (Belfrage and Hauf, 2017). Process tracing was conducted for each site to examine how changes in livelihood dynamics influenced the uptake of RWHI, and vice versa. We applied explaining-outcome process tracing, as we knew the outcome we wanted to investigate (adoption of RWHI), but were unclear about the factors that caused it and wanted to understand them to craft a sufficient explanation for the outcome.

Process tracing involved several key steps. Firstly, the outcome of interest– the adoption of RWHI systems and the integration of RWHI into households' livelihood strategies– was identified. Secondly, evidence was established by collecting and categorizing data from primary and secondary sources, specifically looking at changes in livelihood dynamics and the adoption of RWHI systems in each study site. Thirdly, processes of change were documented by creating a timeline that outlined the sequence of events and relevant outcomes in the two sites. Alternative causal explanations were explored by gathering more data from primary and secondary sources. Then, using process tracing tests such as the straw in the wind, hoop, smoking gun, and doubly decisive, evidence was assessed by analyzing the most compelling explanations and identifying further evidence that would strengthen the causal explanations. Finally, the evidence was tested to evaluate the extent to which the explanations contributed or did not contribute to the outcome of interest.

The comparative case study method, which is often utilized for uncovering causal patterns of explanation, was used alongside process tracing. It helped add structure and focus to gathered information and strengthened the rigor and validity of the research. While process tracing was used as a within-case method that aims to identify and explain causal mechanisms in one particular case, the comparative case study method was used as a cross-case method to explain similarities and differences across cases, identify patterns and generate hypotheses. Combining process tracing with the comparative case study method allowed for a more nuanced and comprehensive understanding of causal mechanisms and contextual factors that influence them (Bennett and Checkel, 2012; Bartlett and Vavrus, 2017).

First, 9 policy documents and 11 project reports by NGOs and research organizations were analyzed to gather development plans, progress, and agricultural-specific reports from the counties where the study sites are situated, and identify relevant RWHI interventions. A systematic literature review of 97 journal articles and 5 academic theses produced information specific to RWHI adoption dynamics, as well as more general information smallholder agricultural production in Kenya. Finally, 41 semi-structures and un-structured interviews, ranging from 45 to 90 min (18 smallholder farmers, 9 employees at NGOs, 3 public officials, 4 researchers, 6 employees at private sector companies, 1 freelance international development consultant) and 1 focus group discussion (4 members of NGOs promoting RWHI adoption) were carried out. The interviews and focus group discussion were mainly carried out in English, but those with less familiarity with English spoke in either Swahili or Akamba, which was translated for the researcher. The line of questioning differed based on who was interviewed, but broadly sought to understand specifics of RWHI adoption in the site on one hand and broader livelihood dynamics on the other. Questions included a. household decision-making around RWHI adoption, including motivating factors, challenges, and desired outcomes; b. the history of RWHI in the area, including the establishment of input companies, public-sector and NGO initiatives, growth of output markets; and c. the history of the site, including in-migration, changes in land and water use, settlement patterns, and seasonal out-migration.

During primary data collection, interviews, observations and notes were transcribed regularly to identify emerging themes and gaps. Due to limited availability of information regarding current levels of household adoption in each site, participants were identified through snowball sampling once familiarity with the sites was established. Participants were selected based on whether they were currently using farm ponds for irrigation purposes or had done so in the last three years. This allowed for closer examination of reasons for adoption and abandonment in the present time. Process tracing and the comparative case study method were helpful in identifying how similarities and differences in the vulnerability context and transforming structures and processes in each site shaped household capital assets, livelihood strategies and levels of adoption. Then, findings were coded, compared between the sites, and categorized into themes to highlight similarities and differences in both historical adoption processes as well as current adoption dynamics. These activities took place over eight months in Kenya, between February 2022 and September 2022.

4. Analysis and findings

The dynamics present in sustainability transitions can be categorized in three phases from the creation of a niche for innovation system development to when the innovation eventually began to affect regime- and landscape-level dynamics. In the following sections, these three phases will be elaborated on for the case of RWHI in Laikipia and Yatta (see Fig. 3).

4.1. Phase 1: landscape developments place stress on livelihood assets and create niche for RWHI innovation

The Laikipia and Yatta Plateaus were inhabited mainly by nomadic tribes (Maasai and Akamba, respectively) before the colonial administration wrested control of these lands in the early 20th century. Settlers used Laikipia and Yatta for agricultural and livestock production, prioritizing infrastructure development that provided water resources and connected the plateaus to nearby towns to facilitate the movement of exports (Ambani, 2015; Winnifred et al., 2021). The growth of the coffee and horticultural industry after WWII served as a further incentive for the colonial administration to invest in physical infrastructure in these two areas (Tiffen et al., 1994; Ambani, 2015). As settlements grew, native communities were relegated to "reserves" and had limited access to natural capital, even as populations in reserves were rising and the need for food and water grew. Since movement was restricted, communities could not move out of the reserves in search of more water as they were once able to Parker (2020).

Migration into Yatta Plateau took place after independence in 1963, when restrictions on movement eased (Parker, 2020). In interviews, farmers reported that the earliest migrants settled on land around the few surface water bodies that existed in Yatta, limiting water access for later arrivals. Most communities turned to agricultural production or off-farm work for income as their

livelihood strategies, abandoning their traditional pastoral livelihoods as there was little room for herds to graze. However, water scarcity due to frequent droughts, limited surface water resources, ill-maintained water reservoirs, and deforestation meant food and water insecurity was common (Rocheleau et al., 1995). By the 1980s, most of the land in Yatta was claimed, leaving little room for further agricultural expansion (Tiffen et al., 1994). One farmer recalled, "During the 1960s and 1970s, people first started settling along rivers and streams, using all the water" (interview number 8). In Laikipia, even though land had been allocated to certain populations post-Independence, it lay largely unused. It was only during the horticultural boom in the 1980s, with foreign demand increasing prices for vegetables, fruits and flowers, that people began to migrate into the area. People worked for large horticultural farms as laborers or outgrowers, or grew horticultural produce to sell to the market themselves (Ulrich, 2014). The rapidly rising population led to an increased demand for water for both domestic and irrigation purposes. Conflicts between the pastoralists that had moved into Laikipia post-independence and the newly settled farming communities over water became commonplace (Ngigi, 2003; Ngigi et al., 2008).

As population densities increased in both Yatta and Laikipia, land and water resources shrunk and households struggled to grow enough food for themselves. Rainfall alone was not enough to sustain agricultural production, particularly for thirstier, high-value horticultural crops. Droughts and dry spells are relatively commonplace (Ngigi, 2003; Mburu et al., 2015; Masindano, 1996). These landscape pressures put pressure on the prevalent agricultural systems, which were no longer able to meet the livelihood needs of the growing population, and drove the need for innovation. Adopting intensification practices, where more yield could be produced from the same amount of land, became increasingly necessary. Farmers in Yatta adopted terracing measures to mitigate soil erosion and improve water absorption, while farmers in Laikipia often turned to unpermitted abstraction from rivers and streams to irrigate their crops. By the 1980s, growing water scarcity in both Laikipia and Yatta led to efforts by NGOs to encourage the adoption of RWHI systems, particularly in Laikipia where conflict over water resources was commonplace (Masindano, 1996; Ulrich et al., 2012). These actors helped pioneer the development of the RWHI innovation at the niche level and generated research and knowledge around RWHI technologies as part of their initiatives. While NGO initiatives did not result in sustained adoption of RWHI (Ngigi, 2003; Wiesmann, 2008), interview responses indicated that they did spread knowledge and awareness of RWHI practices within communities and led to the adoption of rooftop water harvesting to meet domestic water needs.

4.2. Phase 2: new livelihood opportunities arise from landscape pressures and resulting regime changes

In the subsequent decades, several "windows of opportunity" for RWHI to scale emerged due to developments at the landscape level. By the 1980s, large-scale commercial horticultural farms were being established in Laikipia, supported by government policies and priorities that aimed to increase revenue and foreign exchange reserves (Tyce, 2020). The horticultural industry employed smallholder farmers as outgrowers, who benefitted from subsidized provision of inputs, technical advice and support, and more reliable access to markets. Even those farmers not directly employed by the horticultural industry, but who sold directly on the market, benefited from improved access to market information and industry-driven infrastructure development. This lowered transaction costs and helped them tap into horticultural markets as individuals or collectives (Ulrich, 2014; Recha et al., 2018). As one smallholder farmer recalled, "In 1995–6, people started to see the potential in farming, especially contract farming, and it started to take off" (interview number 5). From the 1990s, Water Resource User Associations (WRUAs), the umbrella bodies under which water projects were established, were fairly successful in managing conflicts within communities and between farmers and pastoralists, and distributing enough water from nearby rivers and streams for households' domestic needs in Laikipia (Lesrima, 2019). This increased the time, labor and money available to households and allowed them to consider new livelihood strategies. However, the amount of water allocated to each household was still not enough for irrigation. A farmer recalled, "The Water Resource Users Association was only able to allocate 5.4 m^3 of water to each farm per day, which was not enough to farm well…that's when I started thinking of having my own water storage facility" (Interview number 26).

As the output market continued to grow in Laikipia, many farmers saw RWHI as an investment that could enable greater participation in the horticultural sector rather than an expensive and unreliable way to meet subsistence needs. Large-scale farms, with their access to finance and exposure to international markets, were able to find a way to make water harvesting viable: through the use of ultraviolet resistant polythene sheets, also known as dam liners. Dam liners, developed in Israel, were able to limit the seepage issues that had plagued efforts to carry out RWHI successfully over the last few decades (Ngigi, 2003). The introduction of dam liners, combined with the growth of the horticultural industry, drove the uptake of farm ponds among smallholder farmers in Laikipia. These developments drove the adoption of RWHI technologies from limited use in protected niche spaces to widespread use in commercially oriented farms. Farmers interested in growing high value horticultural crops for the market, and who could afford the initial costs of RWHI (usually from off-farm income), invested in lined farm ponds. Yields for those farmers that used lined farm ponds for irrigation of horticultural crops were many times higher than those who did not (Wachira, 2013). A range of irrigation companies providing RWHI equipment and associated services began entering the market, and competition between them resulted in a more varied and affordable supply of RWHI system components for farmers to choose from. In interviews, farmers observed that the price of dam liners had declined over the years, making adoption easier. RWHI technologies were finding a place among existing actors in the dominant agrifood regime whose livelihoods were geared towards commercial production. As an employee at an irrigation company in the area said, "The demand for dam liners is very high as using a rainwater harvesting system saves costs on pumping water…most farmers who choose to do a farm pond are engaged in commercial production." (Interview number 3).

In Yatta, on the other hand, household water insecurity continues to affect much of the population. Unlike Laikipia, where there is usually enough surface water to meet basic household subsistence needs due to rivers and streams flowing from Mt Kenya, Yatta has comparatively few surface water bodies. Climate change has made dry spells more frequent, affecting the quantity and quality of water

in borewells, reservoirs and canals. These infrastructures have remained under-funded and badly maintained in the decades since independence; while there are some rivers, few are still perennial (Nguta, 1992; Ahmed et al., 2013; Liavoga et al., 2014). In large part because surface water resources were so limited and were barely able to meet domestic needs, large-scale horticultural industries failed to develop. Households utilized their land for rainfed production of staple crops for (mainly) home consumption. One farmer said that "during times of drought, we scale down our activities and only use water for domestic purposes" (Interview number 37). Commercial agricultural production without access to a viable and lucrative output market did not seem to be a sound investment to many smallholder farmers. At the same time, off-farm opportunities were growing, and were a "much more profitable use of their time than producing unsaleable food crops" (Tiffen et al., 1994:32). Both Machakos and Kitui towns, the closest urban areas to Yatta, grew to be some of the biggest urban areas in Kenya and opened up many opportunities for employment. With so many people in the area migrating for work and more children going to school, households were often unable to depend on family labor or even hired labor for subsistence agricultural production, let alone resource and labor-intensive commercial production (Kubbinga, 2012; Kavoi et al., 2016). Since livelihood strategies were not oriented towards intensifying commercial production, assuming the high initial costs of RWHI system adoption did not seem a sound investment and the uptake of RWHI remained limited.

4.3. Phase 3: RWHI adoption reconfigures household assets and contributes to new livelihood strategies and outcomes

In Laikipia, as experience with and exposure to RWHI systems rose in the dominant agrifood regime, farmers began to feel more confident about investing in other technologies that are often part of a RWHI system, such as water pumps and drip kits, to maximize the system's potential. A range of agricultural companies that provided RWHI equipment and associated services had begun entering the market, and competition between them resulted in a more varied and affordable supply of RWHI system components for farmers to choose from. In addition to irrigation companies, community-organized credit and savings cooperatives (SACCOs) in the area also began to facilitate access to RWHI technologies. In Laikipia, men were mainly the ones leading initiatives to adopt RWHI, although a few women took the lead when their husbands were still tied to an off-farm livelihood. The widespread adoption of RWHI by a range of commercial farmers, with operations of all different sizes, shifted dynamics in the dominant agrifood regime. Besides horticultural production, several farmers were also able to pursue other farm-based livelihood strategies specifically related to RWHI, further increasing adoption. For instance, one farmer had dug several farm ponds as part of a model farm, where he taught other farmers on the best ways to carry out RWHI for irrigation. Another farmer charged others for his advice on how to build and maintain farm ponds. Many farmers had long-term plans on how they wanted to continue enhancing their incomes through farm-based activities that were dependent on RWHI: bee-keeping, cultivating fish, or establishing nurseries or greenhouses.

In Yatta, the difficulties of agricultural production, even for subsistence, made households disinclined to substantially invest in agricultural innovations, including RWHI (Ahmed et al., 2013; Kavoi et al., 2016). Most agriculture in the area continued to depend on traditional farm implements such as hoes (Makau et al., 2014). Due to persistent water insecurity, a lack of input and output markets for high-value crops, and relatively less viable hydrological and geographical conditions, communities follow an investment strategy that prioritizes the development of off-farm incomes. Wages from off-farm employment, which provided the cash to buy food off the market, could offset a lack of agricultural production (Rocheleau et al., 1995). Without "windows of opportunity" for the adoption of RWHI opening up at the regime level, RWHI technologies were not able to break out of its protected niche space. Currently, most farm ponds in Yatta are built with the help and funding of NGOs, but since their designs are standardized to make distribution and installation easier within project timelines, they do not always meet specific household needs.

As men in Yatta sought off-farm work and migrated to nearby towns and cities for employment, women grew food for the household. As an NGO worker said, "women are in charge of household farming because men have gone into town to search for jobs" (Interview number 31). NGOs in Yatta concentrate on working with these women to encourage RWHI adoption for domestic food and water security, often through women self-help groups. Water from these farm ponds is mainly used for subsistence production, with women selling some surplus on the market for extra money. Interviews with the women indicated that their husbands did not regard RWHI as a good investment and were reluctant to spend household income on it. One woman remarked, "Most of the RWHI trainings are for women…men will not come because they do not think it is useful" (interview number 28). Indeed, it was observed that few households had invested in technologies to maximize the potential of their RWHI systems, such as drip lines, sprinklers or pumps, and many RWHI systems were in disrepair due lack of funds for maintenance. Discussions with NGO workers revealed that men were uninterested in RWHI because they saw subsistence production (and irrigation) as a women's job and were only interested in agricultural activities with clear income-generating opportunities. One NGO employee said that "men will only come to trainings when there is the potential to earn money" (interview number 16). Cultural factors could have also affected decision-making on investments. The Akamba were traditionally pastoral and had a history of out-migration, while farming was part of the traditional livelihood strategies of the Kikuyu who settled in Laikipia (Tignor, 1976; Winnifred et al., 2021).

5. Discussion

5.1. An agricultural innovation's level of adoption depends on the role of natural capital in livelihood strategies

Environmental factors have a major influence on the direction of agrifood system transformation, as resource abundancies or scarcities can shape investment decisions around innovation adoption (Köhler et al., 2019, Vermunt et al., 2020). However, as Hansen and Coenen (2015) point out, few contributions to the sustainability transitions literature deal explicitly with the role of local natural resource endowments. Integrating the SLF into the MLP provides a means to engage more deeply with ecological implications of

transitions triggered by the introduction of a niche innovation. Due to the SLF's vulnerability context and transforming processes and structures concepts, the combined framework enables an explicit analysis of changing levels of natural capital and how these changes impact transition dynamics. Findings demonstrate that the role of the natural environment in sustainability transitions has to be understood in the context of households trying to navigate a limited and often unreliable asset base to plan for multiple livelihood strategies. People's access to different levels and combinations of assets has a major influence on how they leverage natural resources to pursue desired livelihood strategies, and what their perception is of technological innovations that can contribute to these strategies.

As this case demonstrates, resource scarcity may stimulate investments in innovations. In Laikipia, for example, rising levels of migration to the area, shrinking water resources for irrigation and limited land for expansion prompted farmers to look for ways to maximize the potential of the natural capital they had access to by adopting RWHI. This was especially urgent in Laikipia due to recurring conflict between farmers and pastoralists over water resources. In this context, RWHI provided a means for households to increase water availability for irrigation where surface water resources were rapidly depleting and the demand for lucrative horticultural products was growing. In Yatta, on the other hand, with fewer surface water resources and higher levels of domestic water insecurity, households were less inclined to deepen farm-based livelihood strategies that would increase reliance on water, which limited the use of RWHI. These dynamics indicate that there are upper and lower bounds of natural capital within which adoption of agricultural innovations becomes more likely. Simply put, there needs to be enough natural capital for agricultural innovation adoption to generate relatively higher returns compared to other livelihood strategies that households could potentially pursue. The inclination to adopt innovations to intensify production when agriculture is profitable and land is scarce is already recognized in the field of agricultural development in the Global South (Boyd and Slaymaker, 2000), but has not been paid much attention to in agrifood transitions literature.

5.2. Sustainability transitions can be catalyzed and accelerated when feedback loops are formed between household assets

Transitions governance has long been concerned with catalyzing and accelerating sustainability transitions (which typically take decades) given the seriousness of the sustainability issues at hand. A central question is, under what circumstances can sustainability transitions take place more quickly? (Köhler et al., 2019). The application of the framework integrating the SLF and MLP highlights that when transforming structures and processes at the landscape level reconfigure levels, combinations and sequences of assets available to households, windows of opportunity open up for a niche innovation to maximize the potential of these new configurations and alter regime dynamics. This can result in new livelihood opportunities that engender more sustainable outcomes. These findings improve understanding of how, in any given context, landscape developments affect livelihood opportunities at the regime-level, altered regime dynamics influence the uptake of an innovation, innovation adoption affects assets available to households, and assets give rise to different types of livelihood strategies (Ingram, 2015, 2018; Köhler et al., 2019). Incorporating the SLF lens into the MLP also provides a pathway to begin forming more generalizable knowledge and insights on how place-specificity can catalyze or accelerate of transitions; until recently, much of the research on the geography of transitions has not moved beyond showing that it is important to consider (Hansen and Coenen, 2015).

Findings from this case show that reinforcing feedback loops between physical, social, and human capital at the regime level can increase the likelihood of innovation adoption and accelerate the sustainability transition. In Laikipia, for example, the formation of WRUAs (social capital) helped households attain domestic water security (natural capital), which decreased the amount of time and labor households had to spend on sourcing water (human capital) and enabled engagement in other livelihood strategies incorporating RWHI (financial capital). Similarly, the rise of the horticultural sector increased knowledge and awareness of horticultural production (human capital), facilitated the development of infrastructure (physical capital), and created new livelihood opportunities incorporating RWHI (financial capital). When innovations allow households to tap into a virtuous cycle of multiplying assets that support a particular livelihood strategy, without needing to make additional investments of their own to increase access to each asset, they are more likely to be adopted. These innovations are also more likely to be income generating compared to other innovations not embedded in such feedback loops. With a well-designed RWHI system, households in Laikipia were able to recover high initial costs in a relatively short time. This enabled them to make further improvements that maximized RWHI potential, mainly by adapting the system to current needs and providing necessary maintenance and upkeep. Such strategies helped increase household financial assets, further accelerating the transition.

5.3. The potential for a sustainability transition is influenced by the presence of socio-cultural institutions

Transitions scholars have long engaged with the role of institutions in shaping transition dynamics. In addition to formal visions and policies, informal institutions such as territorially bound values, norms and practices are important to consider (Hansen and Coenen, 2015; Köhler et al., 2019). The framework integrating the SLF and MLP enables assessment of how socio-cultural institutions influence transition dynamics by impacting livelihoods. This extends understanding of the localized social mechanisms and dynamics underpinning transitions and create spaces for analysis of social differences (gender, class, ethnicity). In Yatta, for example, findings showed that gendered institutions contributed to the belief that RWHI systems can only be used for subsistence production, which is perceived as a woman's responsibility. Men, more interested in income-generating innovations, tended to dismiss the potential of RWHI for commercial production and are unwilling to invest money to upgrade system components. Since men usually control household finances, own the titles to the land, and have greater access to loans, their perception of RWHI has probably contributed to lower levels of sustained adoption. Conversely, men's interest in RWHI in Laikipia likely contributed to higher levels of adoption there. The gendered dimension of transforming processes and structures clearly shaped transition pathways and the likelihood of achieving

desired sustainability outcomes (Mutua, 2014; Theis et al., 2018).

A deeper understanding of the role of institutions in forming and maintaining livelihoods would therefore enable more precise assessments on degrees of path dependence, lock-in mechanisms, as well as tensions within the regime. In the Global South, for example, in decisions regarding innovation adoption– in particular, those innovations that require a substantial investment in time, labor or money– households often choose to minimize risks rather than pursue uncertain profits, especially in unstable or low-resource environments susceptible to shocks. Incorporating agricultural innovations in livelihood strategies often exacerbates a household's vulnerability due to increased dependence on market prices, agricultural inputs, infrastructure and even the technology itself (Bridle et al., 2020). To deal with the high levels of risk aversity, the innovation must therefore be able to substantially increase returns from a reliable livelihood opportunity that is aligned with a household "basket" of livelihood strategies without threatening the household's limited asset base. Similar insights could further the creation of policy-relevant scenarios and toolkits that are more suited to prevalent socio-cultural institutions in Global South contexts (Klitkou et al., 2015).

5.4. Further research applications of the inclusion of livelihoods in analysis of sustainability transitions

As discussed earlier, the SLF was formulated to guide development interventions in the Global South and centers the type of livelihoods that development practitioners aim to support. These livelihoods are informal, temporary, and often insecure, dependent to some extent on agricultural or livestock production, and based on household-level priorities. The application of the framework integrating the SLF and the MLP to a case where these types of livelihoods are not dominant may be limited, but it could still offer novel insights on sustainability transitions dynamics in the Global North. After all, decarbonization pathways, which the literature from the Global North tends to focus on, will inevitably involve some winners and losers (Eisenberg, 2019). While some groups may benefit from the creation of new livelihood opportunities in the green energy space, others working in fossil fuel industries may suffer from job losses and limited future prospects (Anderson et al., 2020; Sovacool et al., 2021; Martiskainen et al., 2021). Noting this, the Silesia Declaration of COP24 stated that, "…the just transition of the workforce and the creation of decent work and quality jobs are crucial to ensure an effective and inclusive transition to low greenhouse gas emissions and climate resilient development, and to enhance the public support for achieving the long-term goals of the Paris Agreement" (Just Transition at COP24, 2018).

The concept of "just transitions" arose in the 1970s due to conflicts in the United States between labor unions and policies aimed at promoting new environmental standards. It emphasized that workers' rights need to be protected as economies shift to more sustainable production and consumption patterns. More recently, in response to criticism that transitions literature has downplayed the role of worker agency, there has been growing scholarship that re-engages with grassroots activism and labor unions as forces shaping low-carbon transitions (Wang and Lo, 2021; Morena et al., 2020). For this reason, the framework integrating the SLF and the MLP could be one entry point to better understand how changes in employment opportunities affect transition pathways in the Global North. It could, for example, enhance appreciation of how decarbonization efforts impact different types of households. Certain households may experience unaffordable rises in fuel prices as governments reconfigure policies, institutions and infrastructure in their transition to clean energy, which may alter the makeup of household livelihood assets and strategies. Even well-to-do households that are incentivized to adopt innovations such electric vehicles may be frustrated by changes in payment systems, driving routes, and household infrastructure. Such effects on livelihood opportunities, strategies and assets may engender resistance to sustainability transitions in higher-income countries.

6. Conclusions

Transitions literature does not typically include analysis of livelihoods. However, the inclusion of a livelihoods perspective can improve understanding of the dynamics of sustainability transitions and transition pathways, particularly in the Global South. To deepen insights of the role of livelihoods in sustainability transitions, this paper applies a novel framework that integrates the Sustainable Livelihoods Framework (SLF) from development studies and the Multi-level Perspective (MLP) from transitions literature. This framework was developed to study the case of rainwater harvesting for irrigation (RWHI) in Kenya, a technological innovation aimed at increasing smallholder access to irrigation and decreasing the over-abstraction of surface water resources. Findings demonstrate that livelihood opportunities, assets and strategies played a central role in determining whether the innovation would be adopted and scaled in dominant agrifood systems. In Laikipia, livelihoods are geared towards intensifying on-farm investments, resulting in increasing uptake of RWHI. In Yatta, where livelihoods are based on off-farm activities, adoption of RWHI has stagnated.

Focusing on how livelihoods affect and are affected by transition dynamics helps highlight several features of sustainability transitions in agrifood systems. For one, the potential of an innovation to catalyze sustainability transitions depends on how it alters households' access to natural capital. When access to natural capital is low, and an innovation can maximize available natural resources without putting other assets at risk, it is more likely to be adopted. Second, sustainability transitions can be accelerated when the adoption of an innovation increases access to social, physical, financial and human capital assets. This can take place when landscape developments result in reinforcing feedback loops between different types of capital at the regime-level. Third, informal socio-cultural institutions shaping livelihoods play a major role in determining whether a sustainability transition can take place. Social and cultural norms influence household decision-making around livelihood strategies, demonstrating that profit-driven motivations are rarely the only factor in the adoption of an innovation. The inclusion of livelihoods in analyses of sustainability transitions can also be relevant for the Global North, where decarbonization has major implications for jobs and labor.

This article, with its focus on rural contexts in the Global South, elaborates on the role that livelihoods play in the different phases of an agrifood sustainability transition. Livelihoods are central to the development of a country's cultural, political and economic

institutions and resulting social and environmental outcomes. They constrain or catalyze niche formation, influence regime dynamics and ultimately determine sustainability outcomes. Including a livelihoods perspective can inform policy and development interventions aimed at increasing the adoption of an innovation in a specific context. Ultimately, recognition of the role of livelihoods is fundamental for understanding how a sustainability transition can be achieved and assessing how sustainable and just a transition actually is. It grounds analysis of sustainability transitions in people's everyday strategies for engaging with wider social and environmental systems, as they pursue opportunities for more sustainable livelihoods in the long-term.

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CRediT authorship contribution statement

Radhika Singh: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Visualization. **Jampel Dell'Angelo:** Conceptualization, Writing – review & editing. **Nicholas Oguge:** Supervision, Writing – review & editing. **Collins Odote:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Adato, M., Suseela Meinzen-Dick, R., 2002. Assessing the impact of agricultural research on poverty using the sustainable livelihoods framework. IFPRI, 581-2016-39396.
Ahmed, I., Onwonga, R.N., Mburu, D.M., Elhadi, Y.A., 2013. Evaluation of types and factors influencing adoption of rainwater harvesting techniques in Yatta District, Kenya. Int. J. Educ. Res.
Alinovi, L., D'errico, M., Mane, E., Romano, D., 2010. Livelihoods strategies and household resilience to food insecurity: an empirical analysis to Kenya. Eur. Rep. Dev. 1–52.
Ambani, S.E. 2015. Colonial Urbanization in Kenya: The Case of Nanyuki Town, 1920–1963″. Egerton University, MA in History.
Avelino, F., Wittmayer, J.M., 2016. Shifting power relations in sustainability transitions: a multi-actor perspective. J. Environ. Policy Plann. 18 (5), 628–649.
Avelino, F., 2017. Power in sustainability transitions: Analysing power and (dis) empowerment in transformative change towards sustainability. Environ. Policy Gov. 27 (6), 505–520.
Avelino, F., Grin, J., Pel, B., Jhagroe, S., 2016. The politics of sustainability transitions. J. Environ. Policy Plann. 18 (5), 557–567.
Baldwin, E., Washington-Ottombre, C., Dell'Angelo, J., Cole, D., Evans, T., 2016. Polycentric governance and irrigation reform in Kenya. Governance 29 (2), 207–225.
Baldwin, E., McCord, P., Dell'Angelo, J., Evans, T., 2018. Collective action in a polycentric water governance system. Environ. Policy Govern. 28 (4), 212–222.
Barrett, CB., Benton, TG., Cooper, KA., Fanzo, J., Gandhi, R., Herrero, M., James, S., et al., 2020. Bundling innovations to transform agri-food systems. Nat. Sustain. 3 (12), 974–976.
Bartlett, L., Vavrus, F., 2017. Comparative case studies: an innovative approach. Nordic J. Comp. Int. Educ. 1 (1).
Belfrage, C., Hauf, F., 2017. The gentle art of retroduction: critical realism, cultural political economy and critical grounded theory. Organ. Stud. 38 (2), 251–271.
Bennett, A., Checkel, J.T., 2012. Process tracing: From philosophical roots to best practices. Process Tracing in the Social Sciences, pp. 3–38.
Berkhout, F., et al., 2015. European policy responses to climate change: progress on mainstreaming emissions reduction and adaptation. Reg. Environ. Chang. 15 (6), 949–959.
Bernstein, H., 2010. Class Dynamics of Agrarian Change, 1. Kumarian Press.
Biggs, E.M., et al., 2015. Sustainable development and the water–energy–food nexus: a perspective on livelihoods. Environ. Sci. Policy 54, 389–397.
Binder, C.R., Hinkel, J., Bots, P.W., Pahl-Wostl, C., 2013. Comparison of frameworks for analyzing social-ecological systems. Ecol. Soc. 18 (4).
Boyd, C., Slaymaker, T., 2000. Re-examining the "more people less erosion" hypothesis: special case or wider trend. Nat. Resour. Perspect. 63, 1–6. November.
Bridle, L., Magruder, J., McIntosh, C., Suri, T., 2020. Experimental insights on the constraints to agricultural technology adoption. Agric. Technol. Adopt. Initiative. ATAI.
Bunclark, L., et al., 2018. Understanding farmers' decisions on adaptation to climate change: exploring adoption of water harvesting technologies in Burkina Faso. Glob. Environ. Chang. 48, 243–254.
Chambers, R., Conway, G., 1992. Sustainable rural livelihoods: practical concepts for the 21st century. Inst. Dev. Stud. UK.
Clement, F., 2010. Analysing decentralised natural resource governance: proposition for a "politicised" institutional analysis and development framework. Policy Sci. 43 (2), 129–156.
Cole, D., Epstein, G., McGinnis, M., 2019. The utility of combining the IAD and SES frameworks. Int. J. Commons 13 (1).
De Sherbinin, A., VanWey, LK., McSweeney, K., Aggarwal, R., Barbieri, A., Henry, S., Hunter, LM., Twine, W., Walker, R., 2008. Rural household demographics, livelihoods and the environment. Glob. Environ. Chang. 18 (1), 38–53.
Dell'Angelo, J., et al., 2016. Community water governance on Mount Kenya: an assessment based on Ostrom's design principles of natural resource management. Mt. Res. Dev. 36 (1), 102–115.
Dell'Angelo, J., Rulli, M.C., D'Odorico, P., 2023. Will war in Ukraine escalate the global land rush? Science 379 (6634), 752–755.
Djurfeldt, A.A., 2021. Translocal livelihoods research and the household in the Global South–a gendered perspective. J. Rural Stud. 86, 16–23.
Edouard, M., Krause, D., Stevis, D., 2020. Just Transitions. In Social Justice in a Low-Carbon World.

Edsand, H-E., 2019. Technological innovation system and the wider context: a framework for developing countries. Technol. Soc. 58, 101–150.
Eisenberg, A.M., 2019. Just Transitions. L REV 101.
Hamid, E.B., Ben Hassen, T., Bottalico, F., Berjan, S., Capone, R., 2021. "Acceptance and Adoption of Technologies in Agriculture. Agrofor Int. J. 6 (1), 135–150.
Elzen, B., Geels, FW., Green, K., 2004. System Innovation and the Transition to Sustainability: Theory, Evidence and Policy. Edward Elgar, Cheltenham, UK.
Enfors, E., 2013. Social–ecological traps and transformations in dryland agro-ecosystems: using water system innovations to change the trajectory of development. Glob. Environ. Chang. 23 (1), 51–60.
Feola, G., 2020. Capitalism in sustainability transitions research: time for a critical turn? Environ. Innov. Soc. Trans. 35, 241–250.
Fineschi, R., 2022. Alessandro Mazzone, Per una Teoria Del Conflitto. Scritti 1999-2012. La Città del Sole.
Geels, F.W., Schot, J., 2007. Typology of sociotechnical transition pathways. Res. Policy 36 (3), 399–417.
Geels, F.W., 2002. Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study. Res. Policy 31 (8–9), 1257–1274.
Geels, F.W., 2006. Multi-level perspective on system innovation: relevance for industrial transformation. In: Understanding Industrial Transformation: Views from Different Disciplines. Springer, Dordrecht, Netherlands, pp. 163–186.
Geels, F.W., 2019. Socio-technical transitions to sustainability: a review of criticisms and elaborations of the multi-level perspective. Curr. Opin. Environ. Sustain. 39, 187–201.
Geels, F., Raven, R., 2006. Non-linearity and expectations in niche-development trajectories: ups and downs in Dutch biogas development (1973–2003). Technol. Anal. Strat. Manag. 18 (3–4), 375–392.
Geels, F.W., Turnheim, B., 2022. The Great Reconfiguration. Cambridge University Press.
Goldsmith, P., 2016. Research findings: models of commercial agriculture in Kenya. Futur. Agric.
Gomez, J.T., et al., 2018. Using rainwater for off-season small-scale irrigation in arid and semi-arid areas of Sub-Saharan Africa: key working principles and best practices. Rainwater-Smart Agriculture in Arid and Semi-Arid Areas. Springer, Cham, pp. 9–36 edited by Josep de Trincheria Gomez.
Hall, P., Soskice, D., 2001. An introduction to varieties of capitalism. P. Hall and D. Soskice Varieties of Capitalism: The institutional Foundations of Comparative Advantage: 1–70. Oxford University Press, Oxford.
Hansen, T., Coenen, L., 2015. The geography of sustainability transitions: review, synthesis and reflections on an emergent research field. Environ. Innov. Soc. Trans. 17, 92–109.
Hansen, U.E., Nygaard, I., Romijn, H., Wieczorek, A., Kamp, L.M., Klerkx, L., 2018. Sustainability transitions in developing countries: Stocktaking, new contributions and a research agenda. Environ. Sci. Policy 84, 198–203.
Hebinck, A., Klerkx, L., Elzen, B., Kok, K.P.W., König, B., Schiller, K., Tschersich, J., Mierlo, B., Wirth, T., 2021. Beyond food for thought–Directing sustainability transitions research to address fundamental change in agri-food systems. Environ. Innov. Soc. Trans. 41, 81–85.
El Bilali, H., Hauser, M., Berjan, S., Miseckaite, O., Probst, L., 2017. Rural livelihoods transitions: towards an integration of the sustainable livelihoods approach and the multi-level perspective. In: In International scientific conference Rural Development 2017, pp. 1010–1016.
Ingram, J., 2015. Framing niche-regime linkage as adaptation: an analysis of learning and innovation networks for sustainable agriculture across Europe. J. Rural Stud. 40, 59–75.
Ingram, J., 2018. Agricultural transition: niche and regime knowledge systems' boundary dynamics. Environ. Innov. Soc. Trans. 26, 117–135.
Isgren, E., Ness, B., 2017. Agroecology to promote just sustainability transitions: analysis of a civil society network in the Rwenzori Region, Western Uganda. Sustainability 9 (8), 1357.
Jenkins, K., Sovacool, BK., McCauley, D., 2018. Humanizing sociotechnical transitions through energy justice: an ethical framework for global transformative change. Energy Policy 117, 66–74.
Kaljonen, M., Kortetmäki, T., Tribaldos, T., Huttunen, S., Karttunen, K., Maluf, R.S., Niemi, J., Saarinen, M., Salminen, J., Vaalavuo, M., Valsta, L., 2021. Justice in transitions: widening considerations of justice in dietary transition. Environ. Innov. Soc. Trans. 40, 474–485.
Kavoi, JM., Kamau, GM., Mwangi, JG., 2016. Gender and group dynamics in subsistence agriculture: the case of Kenya. Int. J. Agric. Extension 4 (1), 11–18.
Klitkou, A., Bolwig, S., Hansen, T., Wessberg, N., 2015. The role of lock-in mechanisms in transition processes: the case of energy for road transport. Environ. Innov. Soc. Trans. 16, 22–37.
Köhler, J., Geels, F.W., Kern, F., Markard, J., Onsongo, E., Wieczorek, A., Alkemade, F., Avelino, F., Bergek, A., Boons, F., Fünfschilling, L., 2019. An agenda for sustainability transitions research: state of the art and future directions. Environ. Innov. Soc. Trans. 31, 1–32.
Kubbinga, B., 2012. Road Runoff Harvesting in the Drylands of SSA: Its potential For Assisting Smallholder Farmers in Coping With Water Scarcity and Climate change, Based On Case Studies in Eastern province, Kenya". Vrije University, Amsterdam. MSc Thesis.
Kuhmonen, I., Siltaoja, M., 2022. Farming on the margins: just transition and the resilience of peripheral farms. Environ. Innov. Soc. Trans. 43, 343–357.
Lanari, N., Schuler, R., Kohler, T., Liniger, H., 2018. The impact of commercial horticulture on river water resources in the Upper Ewaso Ng'iro River Basin, Kenya. Mt. Res. Dev. 38 (2), 114–124.
Lesrima, S.S., 2019. Access To Water Resources And Conflicts in the Upper Ewaso Nyiro North River Sub-Basin: A Case Of Laikipia County, Kenya. University of Nairobi. Dissertation.
Levine, S., 2014. How to study livelihoods: bringing a sustainable livelihoods framework to life. Secure Livelihoods Research Consortium. ODI.
Liavoga, B.A., Kathumo, V.M., Onwonga, R.N., Karuku, G.N., Onyango, C.M., 2014. Assessment of trends in land cover and crop type change over two decades in Yatta sub county, Kenya. Int. J. Agric. Forest. Fish. 2, 46–52.
Loewen, B., 2022. Revitalizing varieties of capitalism for sustainability transitions research: review, critique and way forward. Renew. Sustain. Energy Rev. 162, 112432.
Makau, W., et al., 2014. Rain water harvesting for enhanced household water, food and nutritional security: case study of Kitui west, lower Yatta and Matinyani districts, Kenya. J. Environ. Earth Sci. 4 (14), 62–69.
Malesu, M.M. 2007. "Mapping the potential of rainwater harvesting technologies in Africa: a GIS overview and atlas of development domains for the continent and nine selected countries". Vol. 7. World Agroforestry Centre.
Markard, J., Truffer, B., 2008. Technological innovation systems and the multi-level perspective: towards an integrated framework. Res. Policy 37 (4), 596–615.
Markard, J., Hekkert, M., Jacobsson, S., 2015. The technological innovation systems framework: response to six criticisms. Environ. Innov. Soc. Trans. 16, 76–86.
Martiskainen, M., Sovacool, B.K., Hook, A., 2021. Temporality, consumption, and conflict: exploring user-based injustices in European low-carbon transitions. Technol. Anal. Strat. Manag. 33 (7), 770–782.
Masindano, PW., 1996. Security of Subsistence Among Small Scale Migrant Farmers: The Case of Thome settlement Scheme in Laikipia district. University of Nairobi, Kenya. Dissertation.
Mati, B., 2023. Farmer-led irrigation development in Kenya: characteristics and opportunities. Agric. Water Manag. 277, 108105.
Matthews, S., 2004. Post-development theory and the question of alternatives: a view from Africa. Third World Q. 25 (2), 373–384.
Mburu, B.K., Kung'u, J.B., Muriuki, J.N., 2015. Climate change adaptation strategies by small-scale farmers in Yatta District, Kenya. Afr. J. Environ. Sci. Technol. 9 (9), 712–722.
McCord, P., Dell'Angelo, J., Baldwin, E., Evans, T., 2017. Polycentric transformation in Kenyan water governance: a dynamic analysis of institutional and social-ecological change. Policy Stud. J. 45 (4), 633–658.
Mekdaschi, R., Liniger, H., 2013. Water harvesting: guidelines to good practice". Centre Dev. Environ.
Meynard, J.M., Jeuffroy, M.H., Le Bail, M., Lefèvre, A., Magrini, M.B., Michon, C., 2017. Designing coupled innovations for the sustainability transition of agrifood systems. Agric. Syst. 157, 330–339.
Mutua, BN., 2014. Role of Gender On the Growth of the Export Horticulture Industry in Kenya: a Case of Yatta constituency, Machakos County. University of Nairobi. PhD diss.
Mwangi, M., Kariuki, S., 2015. Factors determining adoption of new agricultural technology by smallholder farmers in developing countries. J. Econ. Sustain. Dev. 6 (5).

Natarajan, N., Newsham, A., Rigg, J., Suhardiman, D., 2022. A sustainable livelihoods framework for the 21st century. World Dev. 155, 105898.
Ngigi, S.N., Savenije, H.H.G., Gichuki, FN., 2008. Hydrological impacts of flood storage and management on irrigation water abstraction in upper Ewaso Ng'iro river basin, Kenya. Water Resour. Manag. 22 (12), 1859–1879.
Ngigi, S.N., 2003. What is the limit of up-scaling rainwater harvesting in a river basin? Phys. Chem. Earth Parts A B C 28 (20–27), 943–956.
Nguta, S.M. 1992. "Optimal, farm plans for small holder farmers practising irrigation along Yatta canal of Machakos District, Kenya: a linear programming approach." PhD diss.
Nyamwena-Mukonza, C., 2013. A conceptual framework for implementing sustainable livelihoods and innovation in biofuel production among smallholder farmers. Afr. J. Sci. Technol. Innov. Dev. 5 (3), 180–188.
Parker, J.. 2020. "A wasted eden: colonial water management and ecological change in Kitui, Kenya 1948–63." Les Cahiers d'Afrique de l'Est/The East African Review 55.
Piemontese, L., Castelli, G., Fetzer, I., Barron, J., Liniger, H., Harari, N., Bresci, E., Jaramillo, F., 2020. "Estimating the global potential of water harvesting from successful case studies. Glob. Environ. Chang. 63, 102121.
Pulver, S., Ulibarri, N., Sobocinski, K.L., Alexander, S.M., Johnson, M.L., McCord, P.F., Dell'Angelo, J., 2018. Frontiers in socio-environmental research. Ecol. Soc. 23 (3).
Reed, MS., Podesta, G., Fazey, I., Geeson, N., Hessel, R., Hubacek, K., Letson, D., et al., 2013. Combining analytical frameworks to assess livelihood vulnerability to climate change and analyse adaptation options. Ecol. Econ. 94, 66–77.
Recha, CW. 2018. "Local and regional variations in conditions for agriculture and food security in Kenya.".
Rocheleau, D., Benjamin, P., Diang'a, A., 1995. The Ukambani Region of Kenya. United Nations University.
Scoones, I., 1998. Sustainable Rural Livelihoods: A Framework For Analysis. IDS Working Paper.
Scoones, I., 2009. Livelihoods perspectives and rural development. J. Peasant Stud. 36 (1), 171–196.
Scoones, I., 2015. Sustainable Livelihoods and Rural Development. Practical Action Publishing, Rugby.
Serrat, O., 2017. The sustainable livelihoods approach. In Knowledge Solutions. Springer, Singapore, pp. 21–26 edited by Springer.
Sietz, D., Van Dijk, H., 2015. Land-based adaptation to global change: what drives soil and water conservation in Western Africa? Glob. Environ. Chang. 33, 131–141.
Small, L-A., 2007. The sustainable rural livelihoods approach: a critical review. Can. J. Dev. Stud. Revue 28 (1), 27–38. Canadienne D'études du Développement.
Sovacool, B.K., Turnheim, B., Hook, A., Brock, A., Martiskainen, M., 2021. Dispossessed by Decarbonisation: reducing Vulnerability, Injustice, and Inequality in the Lived Experience of Low-Carbon Pathways. World Dev. 137, 105–116.
Theis, S., et al., 2018. What happens after technology adoption? Gendered aspects of small-scale irrigation technologies in Ethiopia, Ghana, and Tanzania. Agric Human Values 35 (3), 671–684.
Tiffen, M., Mortimore M., and Gichuki F.. 1994. More people, less erosion. Kenyan ed. ACTS Press.
Tignor, R.L., 1976. Colonial Transformation of Kenya: The Kamba, Kikuyu, and Maasai from 1900 to 1939. Princeton University Press.
Tyce, M., 2020. A `private-sector success story'? Uncovering the role of politics and the state in Kenya's horticultural export sector. J. Dev. Stud. 56 (10), 1877–1893.
Ulrich, A., Speranza, C.I., Roden, P., Kiteme, B., Wiesmann, U., Nüsser, M., 2012. Small-scale farming in semi-arid areas: livelihood dynamics between 1997 and 2010 in Laikipia, Kenya. J. Rural Stud. 28 (3), 241–251.
Ulrich, A., 2014. Export-oriented horticultural production in Laikipia, Kenya: assessing the implications for rural livelihoods. Sustainability 6 (1), 336–347.
Vermunt, DA., et al., 2020. Sustainability transitions in the agri-food sector: how ecology affects transition dynamics. Environ. Innov. Soc. Trans. 36, 236–249.
Valverde, R.O., Mesías, P.A., Peris-Blanes, J., 2022. Just transitions through agroecological innovations in family farming in Guatemala: enablers and barriers towards gender equality. In: Environ. Innov. Soc. Trans., 45, pp. 228–245.
van Welie, M.J., Romijn, H.A., 2018. NGOs fostering transitions towards sustainable urban sanitation in low-income countries: Insights from transition management and development studies. Environ. Sci. Policy 84, 250–260.
Van Welie, M.J., Cherunya, P.C., Truffer, B., Murphy, J.T., 2018. Analysing transition pathways in developing cities: the case of Nairobi's splintered sanitation regime. Technol. Forecast. Soc. Chang. 137, 259–271.
Wachira, F.W., 2013. Rainwater Harvesting Practices, Cropping Systems and Productivity in Smallholder Farms in Laikipia Central District. University of Nairobi. PhD diss.
Wang, X., Lo, K., 2021. Just transition: a conceptual review. Energy Res. Soc. Sci. 82, 102291.
Wigboldus, S., Klerkx, L., Leeuwis, C., Schut, M., Muilerman, S., Jochemsen, H., 2016. Systemic perspectives on scaling agricultural innovations: a review". Agron. Sustain. Dev. 36, 46.
Wiesmann, Urs., 2008. Development approaches for sustainable use of water resources in Laikipia, Kenya. Sustainable Regional Development in Rural Africa: Conceptual Framework and Case Studies from Kenya, pp. 217–265.
Williams, S., Doyon, A., 2019. Justice in energy transitions. Environ. Innov. Soc. Trans. 31, 144–153.
Winnifred, M., Nicholus, M., Tanui, P., 2021. From cattle herders to cash earners: the emergence of a migrant wage labour class and its impact on livestock economy among the Akamba of Machakos, Kenya, 1895-1963. Int. J. Hist. Philos. Res. 9 (2), 1–25.
Zhu, T., Ringler, C., Rosegrant, MW., 2019. Viewing agricultural water management through a systems analysis lens. Water Resour. Res. 55 (3), 1778–1791.